

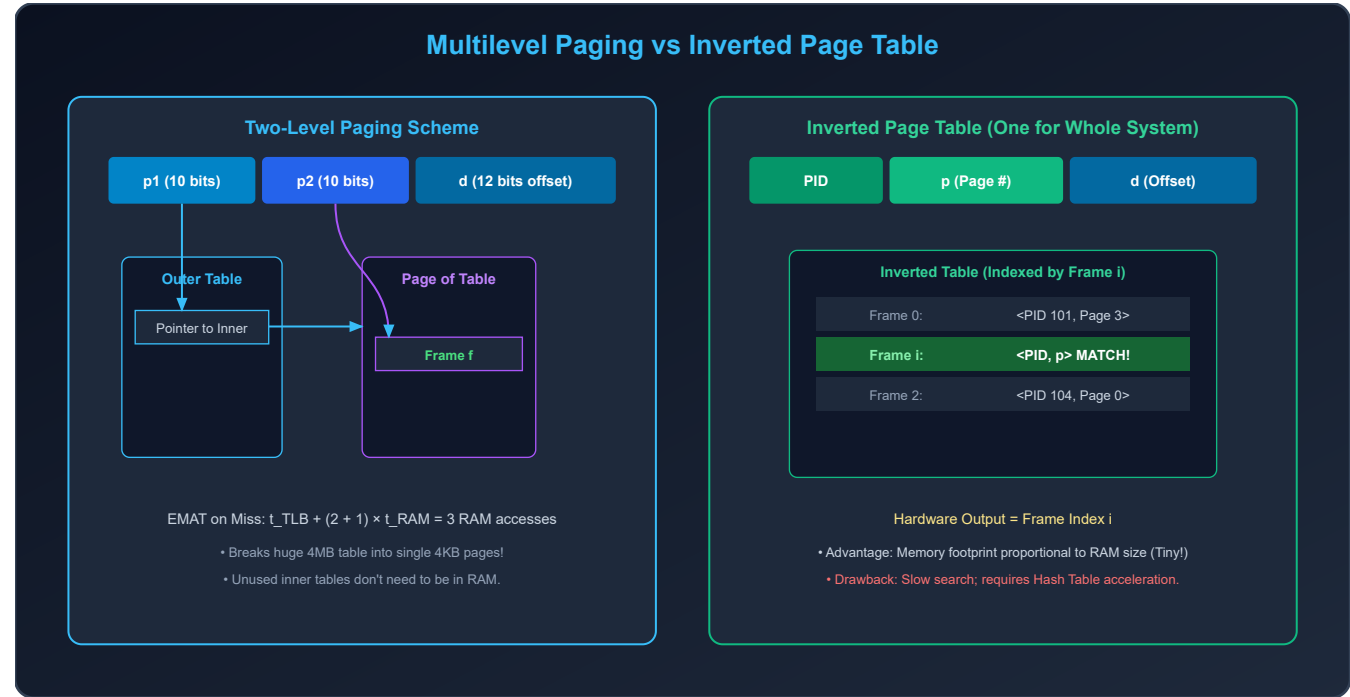
Module 06: Multilevel and Inverted Page Tables

1. Page Table Scaling Problem

32-bit system me 4KB pages ke sath har process ko 4MB ka contiguous page table chahiye. Multilevel paging is samasya ko solve karti hai page table ko khud page karke.

2. Two-Level Paging vs Inverted Page Table

Parameter	Hierarchical (Multilevel) Paging	Inverted Page Table
Number of Tables	Har process ke liye alag multi-level hierarchy	Poore system me strictly EK table
Indexing	Virtual Page Number ke dwara index	Physical Frame Number ke dwara index
Memory Overhead	High virtual space ke sath badhta hai	RAM size ke proportional (Extremely small)
Search Overhead	k-level me (k+1) RAM accesses per miss	Slow linear search (Hash table required)



3. Address Breakdown Formula

Two-Level 32-bit:

Logical Address = [p1 : 10 bits] | [p2 : 10 bits] | [d : 12 bits]

Inverted Address:

Logical Address = <PID, Page Number, Offset> → Matches Frame Index i in table.